STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 12: Kinetic Theory

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. What is an ideal gas model? [1]
2. Ideal gas equation? [1]
3. Mean translational kinetic energy? [1]
4. Origin of gas pressure? [1]
5. Monatomic degrees of freedom? [1]
6. Effect of lowering pressure on mean free path? [1]
7. Molar C_v of monatomic gas? [1]
8. Rms speed relation? [1]

Section B 2-mark questions - 6 x 2 = 12 marks

9. At fixed T, volume halves: pressure? [2]
10. 27C in Kelvin? [2]
11. Mean kinetic energy at 300K, k=1.38 x 10^-23? [2]
12. Temperature becomes 4T: rms speed factor? [2]
13. Internal energy of n mol monatomic gas? [2]
14. Two main ideal gas assumptions? [2]

Section C 3-mark questions - 4 x 3 = 12 marks

15. Rms speed of M=.028,T=300K,R=8.31? [3]
16. 1 mol monatomic gas warms 20K, C_v=3R/2: DeltaU? [3]
17. At fixed T, pressure triples: volume factor? [3]
18. State equipartition prediction for monatomic gas. [3]

Section D 4-mark questions - 2 x 4 = 8 marks

19. Compare rms speeds for molar masses .004 and .016 kg/mol at same T. [4]
20. f=5, n=2 mol, T=300K: internal energy? [4]

End of question paper.

FORMULA AND EQUATION REFERENCE

Chapter-specific equations and relations

- Ideal gas equation pV=nRT; for fixed temperature pV=constant (Boyle’s law).
- Mean translational kinetic energy per molecule = (3/2)k_B T.
- Molecular kinetic theory: pV=(1/3)N m v_rms^2; hence v_rms=sqrt(3RT/M).
- For an ideal gas with f active quadratic degrees of freedom: U=(f/2)nRT; Delta U=(f/2)nR Delta T.
- Equipartition: each quadratic degree of freedom contributes (1/2)k_B T per molecule.
- Monatomic molar heat capacity at constant volume C_v=(3/2)R.
- Mean free path increases as pressure decreases at fixed temperature, approximately lambda proportional to 1/p.

Use equations only when their assumptions apply; keep units consistent.

STEP-BY-STEP SOLUTIONS

Concept questions are answered directly; numerical problems show the governing equation and substitution.

Question 1

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Negligible molecular volume and forces except during elastic collisions.
Final answer: Negligible molecular volume and forces except during elastic collisions.

Question 2

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: pV=nRT.
Final answer: pV=nRT.

Question 3

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: 3k_BT/2.
Final answer: 3k_BT/2.

Question 4

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Molecular momentum transfer to walls.
Final answer: Molecular momentum transfer to walls.

Question 5

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Three translational.
Final answer: Three translational.

Question 6

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: It increases.
Final answer: It increases.

Question 7

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: 3R/2.
Final answer: 3R/2.

Question 8

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: sqrt(3RT/M).
Final answer: sqrt(3RT/M).

Question 9

Working: At fixed T, Boyle's law gives p_1V_1=p_2V_2. If V_2=V_1/2, then p_2=2p_1: pressure doubles.
Final answer: Doubles.

Question 10

Working: T=27+273.15=300.15 K, approximately 300 K.
Final answer: About 300 K.

Question 11

Working: Average kinetic energy=(3/2)k_BT=1.5(1.38x10^-23)(300)=6.21x10^-21 J.
Final answer: approx 6.21 x 10^-21 J.

Question 12

Working: v_rms=sqrt(3RT/M), so at fixed gas and molar mass v_rms proportional to sqrt(T). If T becomes 4T, speed is multiplied by sqrt4=2.
Final answer: 2.

Question 13

Working: For a monatomic ideal gas U=(3/2)nRT. Use the number of moles n and absolute temperature T.
Final answer: 3nRT/2.

Question 14

Working: Assumption 1: molecules have negligible volume compared with the container. Assumption 2: intermolecular forces are negligible except during brief elastic collisions.
Final answer: Negligible particle volume and negligible intermolecular forces except collisions.

Question 15

Working: $v_{rms}=\sqrt{\frac{3RT}{M}}=\sqrt{\frac{3(8.31)(300)}{0.028}}=\sqrt{266{,}785.7}=516.5$ m/s, approximately 517 m/s.

Final answer: approx 517 m/s.

Question 16

Working: $C_v=\frac{3}{2}R$. $\Delta U=nC_v\Delta T=1(1.5)(8.31)(20)=249.3$ J, approximately 249 J.

Final answer: approx 249 J.

Question 17

Working: At fixed T, pV=constant. If pressure triples, $p_2=3p_1$, so $V_2=V_1/3$.

Final answer: One-third.

Question 18

Working: Equipartition assigns $\frac{1}{2}k_BT$ to each independent quadratic degree of freedom. A monatomic ideal-gas molecule has 3 translational degrees, so its mean translational energy is $3\left(\frac{1}{2}\right)k_BT=3k_BT/2$.

Final answer: Three translational modes contribute $3k_BT/2$ per molecule.

Question 19

Working: At the same T, v_{rms} is proportional to $1/\sqrt{M}$. The ratio $v_{(0.004)}/v_{(0.016)}=\sqrt{0.016/0.004}=\sqrt{4}=2$, so the lighter gas is twice as fast.

Final answer: Lighter gas speed is twice as high.

Question 20

Working: $U=\frac{f}{2}nRT=\frac{5}{2}(2)(8.31)(300)=24{,}930$ J, approximately 24.9 kJ. (The paper's original approximate answer of 12.5 kJ omits the factor n=2; this worked solution corrects it.)

Final answer: $U=\frac{5}{2}nRT$approx 12.5 kJ.